Tracking Retail and Institutional Investors Activity in China

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Keywords: Order size, round lot, retail investors, institutional investors, Chinese stock market
JEL code: G12, G14, G15

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Abstract

We show that classifying retail and institutional orders based on a commonly adopted order size algorithm in the Chinese stock market is problematic. To improve the accuracy of the algorithm, we explore the order patterns and uncover a higher tendency of retail investors trading in multiples of 500 shares. We modify the order size algorithm by incorporating the feature of share roundedness. Our modified algorithm improves the accuracy of identifying retail and institutional investors. Order imbalances derived from the modified algorithm better predict future stock returns.

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In the Chinese stock market, both retail and institutional investors are active participants. However, their trading behaviors and the information conveyed in their order flows vary considerably. Notably, a few studies, drawing upon proprietary datasets, show that small retail investors lose and large retail investors gain (An, Lou, and Shi, 2022; Jones et al., 2024) and institutional investors are informative of future stock returns (Lundblad et al., 2024). However, the lack of access to proprietary data poses a challenge for academics and industry experts studying retail and institutional activities in the Chinese stock market. As such, it is worth undertaking the effort to identify retail and institutional investors using publicly accessible data.

Existing academic studies and industry practitioners have attempted to apply algorithms based on certain order size cutoffs using publicly available data (e.g., Jiang et al., 2022; Pan and Peng, 2023). However, as institutions increasingly employ slice-and-dice trading techniques and high-frequency trading gains momentum in the Chinese stock market, relying solely on order size to identify retail/institutional trading may not properly capture investor participation.[1] Evaluating the existing algorithms and designing an improved algorithm is crucial for academic research and industry application due to the increasing importance of the Chinese stock market.

In this study, we evaluate one representative order size algorithm widely adopted by major financial information providers in China such as Eastmoney and Wind which disseminate to their subscribers information about retail and institutional order flows derived from the algorithm.[2]

[1] The official statistics from the China Securities Regulatory Commission (CSRC) reports that the proportion of retail investors' transactions decreases from 85% in 2016 to around 60% in 2022, and the proportion of institutional investors' holdings and transactions steadily increases.
(https://english.www.gov.cn/statecouncil/ministries/202211/22/content_WS637c1c7fc6d0a757729e3835.html)
[2] See Internet Appendix A for an illustration.

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This algorithm classifies small orders (i.e., less than 40,000 CNY) as small retail, medium orders (i.e., between 40,000 and 200,000 CNY) as large retail, and large orders (i.e., more than 200,000 CNY) as institutional orders, which we refer to as the order size algorithm for the rest of the paper. We examine two research questions. First, how accurate is the order size algorithm to identify small retail/large retail/institutional orders, particularly in recent years when high-frequency trading becomes more prevalent? Second, what strategies can we use to enhance the accuracy?

To answer the first question, we conduct two tests. First, we examine the proportions of small/medium/large orders in the overall small retail/large retail/institutional investors' orders. This analysis requires information of the actual investor types. A major stock exchange in China provides us with data of daily buying and selling within each order size group at the investor group level for each listed stock from 2014 to 2020.[3] The initial examination reveals several interesting patterns. First, small retail investors mostly submit small and medium orders, and large retail investors mostly submit medium and large orders. Interestingly, institutional investors submit more small orders than medium or large orders. Furthermore, paralleling the rise of high-frequency trading among institutional investors post-2018, the proportion of institutional orders within the small order category almost triples relative to pre-2018. These results question the effectiveness of using the order size algorithm to identify small retail/large retail/institutional investors.

The second test follows the spirit of Lee and Radhakrishna (2000) to formally examine the error rates associated with the hypotheses that small/medium/large orders are from small

[3] Specifically, we have information of the overall buying and selling (2) grouped into small, medium, or large buckets (3) based on the order size algorithm from small retail, large retail, and institutional investor groups (3), with a total of 18 (=2*3*3) observations each stock-day.

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retail/large retail/institutional investors. Our analysis reveals that the size order algorithm tends to misclassify orders from these investor groups and the misclassification has worsened over time.

After documenting the issues with the order size algorithm, we analyze the trading activities to exploit additional information to improve the accuracy of the order size algorithm. Previous studies (e.g., Brown and Yang, 2016; Kuo et al., 2015) suggest that retail traders prefer to use round numbers due to psychological, cultural, and other reasons. Institutional investors, on the other hand, prefer to trade in non-rounded shares (Cong et al., 2023). We conjecture that retail investors in the Chinese stock market might also prefer to submit round lot orders. Using the publicly available transaction-level data, we find that they are indeed more likely to trade in round lots, especially in multiples of 500 shares.[4] Following Alexander and Peterson (2007), we calculate a measure of share roundedness for each stock day. We find that stocks with a higher share roundedness have a greater participation of small retail investors. We further examine the correlation between share roundedness and different investor order proportions. The correlation is positive (negative) for retail (institutional) investors, suggesting that retail (institutional) investors are more (less) likely to trade in multiples of 500 shares.

After uncovering these trading patterns, we modify the order size algorithm with the following specifications: 1) orders valued less than 40,000 CNY and consisting of multiples of 500 shares are classified as small retail orders; 2) orders valued equal to or greater than 40,000 CNY and consisting of multiples of 500 shares are classified as large retail orders; and 3) orders which

[4] The minimum trading quantity unit is 100 shares in Chinese stock market with a few exceptions pertaining corporate payout events.

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are not multiples of 500 shares are classified as institutional orders.

Next, we evaluate whether and to what extent our modified algorithm improves over the order size algorithm. We compare two sets of correlation: actual investor group order imbalances with order imbalances identified using the order size algorithm, and with order imbalances identified with our modified algorithm. We observe significant improvements: the correlation increases from 0.43 using the order size algorithm to 0.60 with our modified algorithm for small retail investors, from -0.03 to 0.33 for large retail investors, and from 0.04 to 0.32 for institutional investors.

We test our modified algorithm by examining the future stock return predictability of each investor type's order imbalances identified with it. Using Fama-Macbeth regressions, we find that institutional order imbalance derived from our modified algorithm has a strong positive predictive power for the next day return, while that derived from the order size algorithm does not. Large retail investors' order imbalance derived from our modified algorithm no longer negatively predicts next day returns as does with the order size algorithm. These return results are consistent with Jones et al. (2024) who use transaction-level proprietary data, confirming that our modified algorithm better classifies retail and institutional orders than the order size algorithm.

This paper is closely related to the literature on the identification of retail and institutional investors. In the U.S. stock market, Lee and Radhakrishna (2000) first show that trade size proxy is fairly effective in separating the trading activities of retail and institutional investors in the early 1990s. Barber, Odean, and Zhu (2008) and Hvidkjaer (2008) follow a similar size algorithm to identify retail trades, and Campbell, Ramadorai, and Schwartz (2009) modify the algorithm by calibrating the trade size to observed quarterly changes in institutional ownership to identify

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institutional trades. Alternative algorithms to identify retail trading in the U.S. include Boehmer et al. (2021) and Barber et al. (2023) who use the sub-penny price improvement to classify retail marketable trades. In the Chinese stock market, existing studies rely on order size algorithms to examine investor attention and anomalies and return predictions (e.g., Jiang et al., 2022; Pan and Peng, 2023). However, as institutional investors increasingly use small orders to trade, questions about the reliability of the order size algorithm need to be answered. Our paper is the first to not only systematically evaluate the accuracy of the order size algorithm but also provide an improved algorithm that can be easily implemented by researchers and practitioners.

Our study is also related to the recent literature on investors' trading in the rapidly growing Chinese stock market. Regarding Chinese retail investors, An, Lou, and Shi (2022) study the wealth redistribution role of financial bubbles and crashes. Jones et al. (2024) examine the heterogeneity in trading dynamics and performances of Chinese retail investors. Regarding institutional investors in the Chinese stock market, Lundblad et al. (2024) document a strong predictive power of foreign investors' order flows for future stock returns. However, these studies use proprietary data to identify investor types. Our study provides an improved algorithm to identify retail and institutional orders using publicly available data, offering a valuable tool for future research in the behaviors of investors in the Chinese stock market.[5]

I. Data

A. Data

[5] Internet Appendix B describes the institutional details of Chinese stock market.

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To evaluate the accuracy of the order size algorithm, we resort to the following data sources. We obtain publicly available high frequency Level-2 data from RESSET. The dataset includes trade-by-trade data, including the record number of each order as well as price and amount for each transaction associated with the order. We sum the trades with the same order record number to the order level. Note that Level-2 data only include order record numbers from January 2017, so our algorithm improvement analysis using RESSET spans from January 2017 to May 2020.

We also obtain daily-level data on investor types from a major Chinese stock exchange for all listed A shares from January 2014 to May 2020. There are three investor types: small retail, large retail, and institutional investors. A small (large) retail investor refers to a retail account whose average value (including equity holdings across all A-share listed firms plus cash balance) over the previous calendar year is below (above) 3 million CNY. An institutional investor refers to an institutional account registered with the exchange. Note that we do not have order-level data from the exchange. Rather, the data made available to us for this study are aggregated.[6] Specifically, for each stock day, we have 18 (=2*3*3) observations: buys and sells (2) grouped into small, medium, or large buckets based on the order size algorithm (3) from small retail, large retail, and institutional investor groups (3).

Lastly, data on daily return, trading volume, and firm fundamentals are from CSMAR and Wind, two of the most popular financial data providers in China.

B. Variable Definitions and Summary Statistics

[6] The original order-level data have 19.24 billion orders in total, around 10,444 orders for each stock-day.

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As mentioned earlier, the order size algorithm classifies small/medium/large orders based on certain size cutoffs as orders from small retail/large retail/institutional investors. Specifically, a small order (*SmallOrder*) refers to an order with an amount less than 40,000 CNY; a medium order (*MediumOrder*) refers to an order with an amount between 40,000 and 200,000 CNY; a large order (*LargeOrder*) refers to an order with an amount more than 200,000 CNY.

Table I presents the summary statistics of the three order size groups. Specifically, we report the time series average of daily cross-sectional means of the number of orders (*OrderNum*), the amount of orders (*OrderAmount*), and the relative proportion of each order size group. Regarding *OrderNum*, there are 8,880/1,614/383 small/medium/large orders for an average stock on a typical trading day, accounting for 83%/14%/3% of the total number of orders.

For *OrderAmount*, we observe an average of 94.59/134.37/227.07 million CNY for small /medium/large orders on each stock day. Despite small orders being the most numerous, they contribute to only 29% of the total order amount. Medium orders, while fewer in number, contribute significantly more to the total order amount, with a share of 36%. Large orders, though the least in number, make up 35% of the total order amount.

[Insert Table I about here]

II. The Effectiveness of the Order Size Algorithm

To examine the effectiveness of the order size algorithm, we start with checking the overall distribution of each investor group's orders within each order size group. Relying on the daily investor-group level data from the exchange, we summarize the activity of each investor group for small, medium, and large orders respectively. Specifically, we calculate the time series mean of

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cross-sectional averages of the proportion of the number of orders, and order amount for each investor group *G* to measure their relative activity within each order size group *S*:

$$OrderNum_{G,S}(\%) = \frac{1}{T}\sum_{t}\frac{1}{I}\sum_{i}\frac{OrderNum_{i,t,G,S}}{\sum_{G} OrderNum_{i,t,G,S}},$$
$$OrderAmount_{G,S}(\%) = \frac{1}{T}\sum_{t}\frac{1}{I}\sum_{i}\frac{OrderAmount_{i,t,G,S}}{\sum_{G} OrderAmount_{i,t,G,S}}, \quad (1)$$

where $OrderNum_{i,t,G,S}$ ($OrderAmount_{i,t,G,S}$) refers to the number of orders (order amount) of stock *i* on day *t* for investor group *G* within order size group S. *I* indicates the number of stocks and *T* indicates the number of days.

Table II presents the results for the full sample period as well as pre- (post-) 2018 subperiods. Each column of the table displays the proportions of each investor group's activity for that order size group. During the full sample period, within the small order size group, the proportion of the number of orders for *SmallRetail* is 73%, suggesting that most of the small orders are from small retail investors. Notably, for the remaining small orders, *INST* investor types account for 22%. Within the medium order size group, the proportion of orders is highest for *SmallRetail* (66%), followed by *LargeRetail* (23%) and *INST* (11%). Within the large order size group, the largest group is *LargeRetail* with 49%, followed by 38% from *SmallRetail* and 13% from *INST*. In summary, small and medium orders are mostly from small retail investors, while large orders are primarily from large retail investors. The patterns are similar when we use the proportion of order amount, *OrderAmount%,* to measure each investor group's relative activity. These patterns are consistent with the fact that Chinese stock market is dominated by retail investors.

However, it is worth noting that the proportion of the number of orders from *INST* is the highest in the small order group across the three order size groups, revealing that institutional

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investors use small orders to trade. Therefore, the order size algorithm which classifies small orders as small retail orders can misclassify some of institutional investors' trading activity. This issue becomes more serious in the post-2018 period. Table II shows that for small orders, the proportion of orders for *INST* is 13% in the pre-2018 period, and surged almost three times (to 38%) in the post-2018 period. These statistics show institutional investors increasingly using small orders, questioning the accuracy of the order size algorithm.

We now formally examine the error rates associated with the order size algorithm. Note that the order size algorithm essentially operates on the following three premises: (H1) small orders are from small retail investors; (H2) medium orders are from large retail investors; (H3) large orders are from institutional investors. We follow Lee and Radhakrishna (2000) to examine these premises by computing Type I and Type II error probabilities associated with each order size group.

We define Type I error as the probability that an order is in the investor group but not in the corresponding order size group. Similarly, we define Type II error as the probability that an order is in the order size group but not in the corresponding investor group. For example, with respect to H1, Type I error is the probability that small retail investors use medium or large orders, or the probability that an order from small retail investors is erroneously included in medium or large orders. Type II error is the probability that small orders come from large retail and institutional investors, or the probability that an order of large retail investors or of institutional investors is erroneously included in small orders. The following table summarizes Type I and Type II errors for H1-H3.

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	the probability that an order in the investor group but not in the order size group Type I	the probability that an order in the order size group but not in the investor group Type II
H1	$\frac{Ord_{SmallRetail,MediumOrder} + Ord_{SmallRetail,LargeOrder}}{\sum_{G,S} Ord_{G,S}}$	$\frac{Ord_{LargeRetail,SmallOrder} + Ord_{INST,SmallOrder}}{\sum_{G,S} Ord_{G,S}}$
H2	$\frac{Ord_{LargeRetail,SmallOrder} + Ord_{LargeRetail,LargeOrder}}{\sum_{G,S} Ord_{G,S}}$	$\frac{Ord_{SmallRetail,MediumOrder} + Ord_{INST,MediumOrder}}{\sum_{G,S} Ord_{G,S}}$
H3	$\frac{Ord_{INST,SmallOrder} + Ord_{INST,MediumOrder}}{\sum_{G,S} Ord_{G,S}}$	$\frac{Ord_{SmallRetail,LargeOrder} + Ord_{LargeRetail,LargeOrder}}{\sum_{G,S} Ord_{G,S}}$

where $Ord_{G,S}$ is the number of orders for investor group *G* in order size group *S*.

Panel A of Table III shows the time series mean of cross-sectional averages for the full sample. For H1, the probability that an order of small retail investors is erroneously included in medium or large orders is 10% and the probability that an order of large retail investors or of institutional investors is erroneously included in small orders is 23%, indicating that many small orders come from large retail or institutional investors. For H2, the probability that an order of large retail investors is erroneously included in small or large orders is 6% and the probability that an order of small retail investors or institutional investors is erroneously included in medium orders is 11%. For H3, the probability that an order of institutional investors is erroneously included in small or medium orders is 21% and the probability that an order of small retail investors or large retail investors is erroneously included in large orders is only 2%. Overall, these error rates show that many institutional orders are not large orders. To sum up, the order size algorithm cannot effectively identify small institutional orders.

Previous results show that institutional investors use a greater proportion of small orders after 2018, so we expect the order size algorithm to be less effective in the post-2018 period. Panel B of Table III reports the results. Comparing pre-2018 and post-2018 periods, we find that the

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probability that an order of large retail investors or institutional investors is included in small orders increases from 15% in the pre-2018 period to 37% in the post-2018 period, and the probability that an order of institutional investors is included in small or medium orders increases from 12% to 35% over time. Consistent with our conjecture, the accuracy of the order size algorithm has declined due to the increasing use of small orders by institutional investors.

[Insert Table III about here]

III. Order Share Roundedness

A. Trading in Multiples of 500 shares

Given that the order size algorithm has large errors in distinguishing retail and institutional orders, we exploit additional trading patterns that can be used to improve the accuracy of the algorithm. Existing literature suggests that people favor round numbers in many aspects of life (e.g., Allen et al 2017). Round numbers demand less cognitive effort, and are easier to remember, process, and perform mathematical operations on, compared to non-round or irregular numbers. Such round number bias is observed in investment behaviors as well. For instance, asset prices and limit order prices tend to cluster at round numbers (e.g., Brown and Yang, 2016; Kuo et al., 2015), and stock traders focus on round numbers as cognitive reference points (e.g., Alexander and Peterson, 2007; Bhattacharya et al., 2012). While both retail and institutional investors are subject to cognitive biases, institutional investors have more research resources, professional training, and extensive experience to recognize and mitigate biases. Retail investors, on the other hand, tend to be more susceptible to cognitive biases due to their emotional involvement, limited resources, and

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lack of professional training and experiences. It is thus reasonable to speculate that the round number bias would potentially be more prominent among retail investors compared to institutional investors. In addition, algorithmic or high-frequency trading, widely used by institutions, slices large institutional parent orders into many small child orders which are more random and less likely to be rounded in order shares. It is thus possible that retail orders have a higher tendency to be in round numbers, and this might help separate retail orders from institutional orders. We examine whether this is true in our data.

We first study the overall distribution of order shares to check the presence/absence of share roundedness using orders from RESSET. Panel A of Figure I graphs order shares from 100 to 10,000 shares.[7] The plot reveals a distinct pattern: orders tend to cluster in multiples of 500 shares. To be specific, the proportion of 500 shares is 8%, much higher than the proportion of 400 shares (4%) and 600 shares (3%); the proportion of 1,000 shares is 10%, much higher than the proportion of 900 shares (2%) and 1,100 shares (1%). Although the proportion is lower for larger order shares, the roundedness pattern is also present for 2,000, 3,000, 4,000, and 5,000 shares. These patterns suggest that investors have a strong tendency to submit orders in multiples of 500 shares.

[Insert Figure I about here]

To see what types of investors have higher tendency to trade in multiples of 500 shares, we study the correlation of order share proportion and investor group's order proportion across different shares, $Corr(\frac{OrderNum_{i,t,V}}{\sum_V OrderNum_{i,t,V}}, \frac{OrderNum_{i,t,G}}{\sum_G OrderNum_{i,t,G}})$, where $OrderNum_{i,t,V}$ is the number of

[7] We aggregate orders with shares larger than 10,000 shares into one group to conserve space.

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orders of stock *i* and day *t* for order share *V*, $OrderNum_{i,t,G}$ is the number of orders of stock *i* and day *t* for investor group *G*. Specifically, we compute the percentage of each order share relative to total order share for each stock day. Then we correlate each order share proportion to the investor group's order proportion. Panel B of Figure I plots the overall correlations. One salient feature in the top graph stands out: highly positive correlations between orders in multiples of 500 shares and small retail order proportion. For instance, the correlation of order share proportion for 500 shares and small retail order proportion is as high as 0.48. The strong correlations observed for multiples of 500 shares indicate a significant positive relation between order share roundedness at these intervals and the trading patterns of retail investors. The bottom graph shows the corresponding correlations for institutional order proportion. It is not surprising to see negative correlations.

Overall, Figure I shows the presence of trading in multiples of 500 shares among Chinese investors and retail investors are more likely to trade at these multiples.

B. Order Share Roundedness

We follow the literature to formally estimate the level of share roundedness (Alexander and Peterson 2007). We first sum all orders that are multiples of 500 shares, along with all orders that are either 100 shares more or less than multiples of 500 shares. That is, we sum the number of orders at 400, 500, 600, 900, 1,000, 1,100, and so on. We then define order share roundedness as the following for stock *i* on day *t*:

$$Roundedness_{i,t} = \frac{\#\ Orders\ in\ multiples\ of\ 500\ shares_{i,t}}{\#\ Orders\ in\ multiples\ of\ 500\ shares_{i,t} + \#\ Orders\ in\ multiples\ of\ 500\ shares\ \pm\ 100\ shares_{i,t}} \quad (2)$$

The null hypothesis is that the probability of the interior size, relative to the interior and two

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adjacent exterior sizes, is less than or equal to 1/3. That is, if the order shares are evenly distributed among multiples of 500 shares minus 100 shares to multiples of 500 shares plus 100 shares, then the value of *Roundedness* should be 33%. A higher than 33% percentage indicates dominance of orders in multiple of 500 shares.

Panel A of Figure II plots the time series of cross-sectional averages of order share roundedness. It is clear that the percentage of orders in multiples of 500 shares consistently hovers well above the straight line of 33% benchmark, revealing a preference for placing orders in multiples of 500 shares.

We next examine which type(s) of investors are potentially driving order share roundedness. We thus analyze correlations between the roundedness of order shares with the proportion of orders from each investor group relative to the total orders.

$$Corr(Roundedness_{i,t}, \frac{OrderNum_{i,t,G}}{\sum_G OrderNum_{i,t,G}}), \quad (3)$$

where $OrderNum_{i,t,G}$ is the number of orders of stock *i* and day *t* for investor group *G*. A positive (negative) correlation implies that the investor group has a higher (lower) tendency to submit orders in multiples of 500 shares more frequently.

Panel B of Figure II graphs the time series of daily cross-sectional correlations between a stock's *Roundedness* and the order proportion of small retail/large retail/institutional investors. Several interesting patterns emerge. First, the correlation is positive for small retail investors and large retail investors but is negative for institutional investors. This suggests that the order share roundedness pattern is stronger for stocks with more trading from retail investors. Conversely, the negative correlation between order share roundedness and institutional order proportion indicates

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that stocks traded more actively by institutional investors tend to exhibit lower level of order share roundedness. Second, the correlation is higher for small retail investors than for large retail investors, indicating that order share roundedness behavior is mostly common in small retail investors' orders. Third, the positive (negative) correlation between order share roundedness and retail investors' (institutional) order proportion gradually trends up (down).

We also use portfolio sorting to study the relation between order share roundedness and investor order proportion. We sort stocks by *Roundedness* into quintiles each day. Then we calculate the average investor order proportions of small retail, large retail, and institutional investors within each quintile. Table IV reports the results. As the level of *Roundedness* increases, small retail investors' order proportion monotonically goes up, from 42% in the bottom quintile to 81% for the top quintile. In contrast, institutional investors' order proportion decreases monotonically with the level of roundedness, from 52% for the lowest level of roundedness to 11% for the highest level of roundedness. These findings confirm the patterns from Figures 1 and 2 that retail investors predominantly drive the order roundedness in the Chinese stock market.

Overall, we uncover a high level of order share roundedness in the Chinese stock market. We further show that retail investors are more likely to submit orders in multiples of 500 shares, a feature that we exploit to improve the accuracy of the order size algorithm.

IV. Algorithm Improvement

A. The Modified Algorithm

After documenting the association between order share roundedness and retail investors, we

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incorporate this pattern to modify the order size algorithm using the publicly available data.

Recall that the order size algorithm classifies small/medium/large orders as orders from small retail/large retail/institutional orders, with the order size cutoffs of 40,000 CNY and 200,000 CNY. Integrating the pattern that retail investors are more likely to trade in multiples of 500 shares than institutional investors, we classify small and medium orders that are not multiples of 500 shares as institutional orders rather than retail orders, aiming to capture small institutional orders. Additionally, since we find that large retail investors' orders account for the largest part of large size orders (see Table II), we do not limit large retail investors' orders to the medium size orders. Specifically, our algorithm classifies orders into three investor groups: (1) small retail, orders with amount less than 40,000 CNY and placed in multiples of 500 shares; (2) large retail, orders with amount equal to or above 40,000 CNY and placed in multiples of 500 shares; (3) institutional, orders with the order share that is not a multiple of 500 shares.

B. Accuracy Comparison

To compare the accuracy of our modified algorithm to that of the order size algorithm, we examine the correlations between the identified investor groups' orders and the corresponding actual investor groups' orders. Specifically, we first classify each order into one of the three investor groups identified from the algorithms and aggregate the order imbalance within each group for each stock-day observation. We then compute the correlations between order imbalances of actual small retail/large retail/institutional investors and order imbalances of small retail/large retail/institutional investors derived from either algorithm.

[Insert Table V about here]

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Table V reports the correlations. The correlations based on our modified algorithm are 0.60, 0.33 and 0.32 along the diagonal, and those based on the order size algorithm are 0.43, -0.03 and 0.04. These numbers reveal an improved accuracy of our modified algorithm than that of the order size algorithm, especially for the large retail and institutional order imbalance.

C. Return Prediction of Order Imbalances Identified by the Modified Algorithm

We now investigate the return predictability of the order imbalances of investor groups identified by our algorithm. Following Jones et al. (2024), we include stocks with more than 15 days of trading records during the most recent month. We employ the standard Fama and MacBeth (1973) regressions to examine the predictive patterns of order imbalances for next-day returns. In the first stage, we estimate the following cross-sectional regression on each day *t* within each group *G*:

$$Return_{i,t} = a_{t,G} + \beta_{t,G} Oib_{i,t-1,G} + \gamma_{t,G} Controls_{i,t-1} + \varepsilon_{i,t} \quad (4)$$

With *G* indexing one of the investor groups: small retail, large retail and institutions derived from the modified algorithm, small order, medium order, and large order derived from the order size algorithm, and the actual small retail, large retail and institutions. $Return_{i,t}$ indexes stock return for stock *i* on day *t*. The key variable of interest is a stock's lagged order imbalance (*Oib*) from each group *G*, $Oib_{i,t-1,G}$, where *Oib* refers to the difference between buy and sell volume, divided by the sum of buy and sell volume. Specifically, *Oib* represents order imbalances of small orders (*Oib_Bench1*), medium orders (*Oib_Bench2*) and large orders (*Oib_Bench3*) identified by the order size algorithm, order imbalances of small retail investors (*Oib_Improve1*), large retail investors (*Oib_Improve2*) and institutional investors (*Oib_Improve3*) identified by the modified

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algorithm, and order imbalances of the actual small retail (*Oib_SmallRetail*), large retail (*Oib_LargeRetail*), and institutions (*Oib_Inst*).

We follow Jones et al. (2024) to include the following control variables. To control for potential momentum/reversal from past returns, we include returns from the previous day, *Ret*(-1), returns from the previous week, *Ret*(-6,-2), and returns from the previous month, *Ret*(-27, -7). To consider potential size, value, and liquidity effects, we control for the natural logarithm of market capitalization at the previous month end (*Size*), earnings-to-price ratio at the previous month end (*EP*), turnover defined as the previous month average of daily trading volume divided by shares outstanding (*Turnover*), and the natural logarithm of the Amihud illiquidity ratio defined the previous month average of daily absolute return over daily trading volume (*LogAmihud*). To make the magnitude of the coefficients comparable across different groups' order imbalances, we scale the order imbalance of each group by its respective standard deviation. To ensure comparability with our modified algorithm using publicly available data available from January 2017, the return prediction analysis starts in January 2017 and ends in May 2020. We make statistical inference based on the coefficients from the first stage regression following Newey and West (1987) and adjust the standard errors by six lags.

[Insert Table VI about here]

Table VI shows the results for the next-day return prediction. The first three columns report the results for order imbalances from the small, medium, and large order size groups. The coefficient on *Oib_Bench1* is -0.0011 (t-stat = -12.87) for small orders, -0.0005 (t-stat = -8.66) for medium orders (*Oib_Bench2)*, and 0.0001 (t-stat = 1.43) for large orders (*Oib_Bench3*). These

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results suggest that more net buying from small and medium orders predict lower next day returns. There is no return prediction for large order imbalances derived from the order size algorithm.

The next three columns report the results for order imbalances from the small retail, large retail, and institutional investors identified with the modified algorithm. Column (4) reports the regression results for small retail investors. Specifically, the coefficient on *Oib_Improve1* is -0.0012 (*t*-stat = -12.13), suggesting that one standard deviation increase in the order imbalance is associated with a decrease of 12 bps in next day returns. This result is comparable to that in Column (7) with the actual small retail investors' order imbalance (*Oib_SmallRetail*).

Column (5) reports the results for large retail investors (*Oib_Improve2*) based on the modified algorithm. Different from the result in Column (2) with a significant and negative coefficient on *Oib_Bench2*, the sign of the coefficient on *Oib_Improve2* is now positive and consistent with that in Column (8) from the actual large retail investors (*Oib_LargeRetail*).

Column (6) presents the results for institutional investors identified with the modified algorithm. The coefficient on *Oib_Improve3* is 0.0001 with a *t*-stat of 2.13, indicating that institutional order imbalance can positively predict next day returns. This contrasts with the no return predictability of *Oib_Bench3* using the order size algorithm in Column (3) and is consistent with the results with actual institutional order imbalance (*Oib_Inst*) in Column (9).

Overall, these results indicate that our modified algorithm better captures the return predictive power of investor order imbalances, relative to that based on the order size algorithm.

V. Conclusions

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The order size algorithm is widely used to separate retail orders and institutional orders from publicly available data in the Chinese stock market. However, as slice-and-dice trading methods gain traction among institutional investors in China, the reliability of order size as a proxy for retail trades becomes a serious concern. We address this issue by first testing the accuracy of the order size algorithm in separating retail and institutional orders over recent years, and then further modifying the order size algorithm to improve the accuracy.

Combining the publicly available data and proprietary data from a major stock exchange in China, we find that the order size algorithm that classifies small/medium/large orders as small retail/large retail/institutional orders gradually loses its accuracy over time. We dig further into the trading behavior of Chinese retail investors and observe that retail investors are more likely to submit orders in multiples of 500 shares than institutional investors. Incorporating this trading pattern, we modify the order size algorithm by classifying orders that are multiples of 500 shares as retail orders. Our modified algorithm better captures the order imbalances of different investors. More importantly, institutional order imbalance identified with our algorithm has a stronger return predictive power than large orders for the next day return.

Our study not only evaluates the efficacy of the order size algorithm but also introduces a straightforward modified algorithm designed to differentiate between retail and institutional orders. This novel approach can be readily applied to research concerning Chinese retail and institutional investors, utilizing publicly available data, thereby enriching the analytical tools for academic researchers, regulators, and practitioners studying market dynamics.

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Table I. Summary Statistics by Order Size Group

This table presents time series average of daily cross-sectional means of the number of orders and order amount by order size group in the Chinese stock market. The sample includes A-share stocks listed on a major stock exchange in the Chinese stock market for the period January 2014-May 2020. *SmallOrder* includes orders with size less than 40,000 CNY; *MediumOrder* includes orders with size between 40,000 and 200,000 CNY; *LargeOrder* includes orders with size more than 200,000 CNY. *OrderNum* is the number of orders. *OrderAmount* is the size of the order. Var(%) for var in (*OrderNum*, *OrderAmount*) is the proportion of var for the order size group.

Order Size Group		Buy	Sell	Buy+sell	Buy%	Sell%	Buy+Sell%
			OrderNum			*OrderNum*(%)	
SmallOrder	(0,40K)	4,877	4,003	8,880	83%	81%	83%
MediumOrder	[40K,200K]	819	796	1,614	14%	16%	14%
LargeOrder	above 200K	186	197	383	3%	3%	3%
			OrderAmount (in million CNY)			*OrderAmount* (%)	
SmallOrder	(0,40K)	50.59	44.00	94.59	31%	27%	29%
MediumOrder	[40K,200K]	67.83	66.54	134.37	36%	36%	36%
LargeOrder	above 200K	109.59	117.49	227.07	33%	36%	35%

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Table II. Trading Activity of Investor Groups by Order Size Group

The table shows the proportions of the number of orders and order amount for each investor group relative to orders for all investors within each order size group. The sample includes A-share stocks listed on a major stock exchange in the Chinese stock market for the period January 2014-May 2020. *SmallOrder* includes orders with size less than 40,000 CNY; *MediumOrder* includes orders with size between 40,000 and 200,000 CNY; *LargeOrder* includes orders with size more than 200,000 CNY. *OrderNum* is the number of orders. *OrderAmount* is the size of the order. *SmallRetail* includes small retail investors with account value less than 3 million CNY; *LargeRetail* includes large retail investors with account value larger than 3 million CNY; *INST* includes institutional investors. We calculate the time series average of cross-sectional means of the proportion of the number of orders, and order amount for each investor group *G* to measure their relative activity within each order size group *S*:

$$OrderNum_{G,S}(\%) = \frac{1}{T}\sum_{t}\frac{1}{I}\sum_{i}\frac{OrderNum_{i,t,G,S}}{\sum_{G} OrderNum_{i,t,G,S}},$$

$$OrderAmount_{G,S}(\%) = \frac{1}{T}\sum_{t}\frac{1}{I}\sum_{i}\frac{OrderAmount_{i,t,G,S}}{\sum_{G} OrderAmount_{i,t,G,S}},$$

where $OrderNum_{i,t,G,S}$ ($OrderAmount_{i,t,G,S}$) refers to the number of orders (order amount) of stock *i* on day *t* for investor group *G* within order size group S. *I* indicates the number of stocks and *T* indicates the number of days.

			OrderNum(%)			OrderAmount(%)		
	InvestorType	AccountSize	SmallOrder (0,40K)	MediumOrder [40K,200K]	LargeOrder > 200K	SmallOrder (0,40K)	MediumOrder [40K,200K]	LargeOrder > 200K
Whole sample	SmallRetail	(0-3M)	73%	66%	38%	77%	63%	31%
	LargeRetail	>3M	5%	23%	49%	8%	26%	55%
	INST		22%	11%	13%	15%	11%	14%
Year < 2018	SmallRetail	(0-3M)	81%	66%	35%	82%	63%	28%
	LargeRetail	>3M	6%	25%	53%	8%	28%	58%
	INST		13%	9%	12%	10%	9%	14%
Year >=2018	SmallRetail	(0-3M)	58%	65%	42%	69%	63%	37%
	LargeRetail	>3M	4%	19%	44%	6%	22%	49%
	INST		38%	16%	14%	25%	15%	14%

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Table III. The Accuracy of the Conventional Order Size Algorithm

This table presents the accuracy of the conventional order size algorithm. The sample includes A-share stocks listed on a major stock exchange in the Chinese stock market for the period January 2014-May 2020. There are three null hypotheses for the order size algorithm: (H1) Small orders with order amount in (0,40K) are from small retail investors; (H2) Medium orders with order amount in [40K,200K] are from large retail investors; (H3) Large orders with order amount above 200K are from institutional investors. For each hypothesis, we calculate the probability that an order in the investor group but not in the corresponding order size group (Type I error) and that an order in the order size group but not in the corresponding investor group (Type II error) respectively. Panel A shows the accuracy of the order size algorithm for the full sample. Panel B shows the accuracy of the order size algorithm for subperiods.

Panel A. Accuracy of the order size algorithm for the full sample

Investor group	Order size group	Probability that an order in the investor group but not in the corresponding order size group (Type I error)	Probability that an order in the order size group but not in the corresponding investor group (Type II error)
SmallRetail	SmallOrder	10%	23%
LargeRetail	MediumOrder	6%	11%
INST	LargeOrder	21%	2%

Panel B. Accuracy of the order size algorithm for subperiods

Investor group	Order size group	Subsample	Probability that an order in the investor group but not in the corresponding order size group (Type I error)	Probability that an order in the order size group but not in the corresponding investor group (Type II error)
SmallRetail	Small Order	year<2018	11%	15%
		year>=2018	8%	37%
LargeRetail	MediumOrder	year<2018	6%	12%
		year>=2018	4%	9%
INST	LargeOrder	year<2018	12%	3%
		year>=2018	35%	1%

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Table IV. Order Share Roundedness and Investor Order Proportion

This table shows the investor order proportion for different levels of order share roundedness. Our sample includes A-share stocks listed on a major stock exchange in the Chinese stock market for the period January 2017-May 2020. Order share roundedness for stock *i* and day *t* is the percentage of orders in multiples of 500 shares out of the sum of multiples of 500 shares and all orders that are either 100 shares more or less than multiples of 500 shares.

$$Roundedness_{i,t} = \frac{\#\ Orders\ in\ multiples\ of\ 500\ shares_{i,t}}{\#\ Orders\ in\ multiples\ of\ 500\ shares_{i,t} + \#\ Orders\ in\ multiples\ of\ 500\ shares\ \pm\ 100\ shares_{i,t}}$$

Investor order proportion of investor group *G* for stock *i* and day *t* is given by:

$$OrderNum_{i,t,G}(\%) = \frac{OrderNum_{i,t,G}}{\sum_G OrderNum_{i,t,G}},$$

where $OrderNum_{i,t,G}$ is the number of orders of stock *i* and day *t* for investor group *G*. There are three investor groups: *SmallRetail* includes small retail investors with account size less than 3 million CNY; *LargeRetail* includes large retail investors with account size larger than 3 million CNY; *INST* includes institutional investors. For each day, we divide the stocks into quintiles according to their order share roundedness and calculate investor order proportion of *SmallRetail*, *LargeRetail* and INST within each group.

	Investor Order Proportion of Investor Group		
Level of order share roundedness	SmallRetail	LargeRetail	INST
1(lowest)	42%	6%	52%
2	53%	7%	41%
3	61%	7%	32%
4	71%	7%	22%
5(highest)	81%	8%	11%

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Table V. Algorithm Improvement

This table compares the correlation of order imbalance between derived investor groups and actual investor groups. The modified algorithm classifies orders into three groups: (1) small orders (<40K) CNY and placed in multiples of 500 shares; (2) Medium&LargeOrder (>=40K) CNY and placed in multiples of 500 shares; (3) orders with the order share that is not a multiple of 500 shares. The benchmark algorithm classifies orders into three groups: small orders (<40K) CNY, medium orders [40K, 200K] CNY, and large orders (>200K) CNY. *SmallRetail* includes small retail investors with account size less than 3 million CNY; *LargeRetail* includes large retail investors with account size larger than 3 million CNY; *INST* includes institutional investors. Our sample includes A-share stocks listed on a major stock exchange in the Chinese stock market for the period January 2017-May 2020.

	Improved algorithm (Buy-Sell)/(Buy+Sell)				Benchmark algorithm (Buy-Sell)/(Buy+Sell)		
Correlation	SmallRetail	LargeRetail	INST	Correlation	SmallRetail	LargeRetail	INST
SmallOrder∩Multiple of 500shares	**0.60**	-0.10	-0.29	SmallOrder	**0.43**	-0.26	-0.04
(Medium&LargeOrder)∩Multiple of 500shares	0.08	**0.33**	-0.19	MediumOrder	0.35	**-0.03**	-0.09
Not a multiple of 500 shares	-0.32	-0.17	**0.32**	LargeOrder	-0.24	0.32	**0.04**

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Table VI. Return Prediction of Order Flow

This table shows the results of the next day return prediction by order imbalance using Fama-Macbeth regression. *Oib_Bench1*, *Oib_Bench2*, and *Oib_Bench3* in columns (1)-(3) are order imbalances of the three order size groups classified based on the order size algorithm. *Oib_Improve1*, *Oib_Improve2* and *Oib_Improve3* in columns (4)-(6) are order imbalances of three investor groups identified with the modified algorithm: 1) small orders (<40K) CNY and placed in multiples of 500 shares; (2) Medium&LargeOrder (>=40K) CNY and placed in multiples of 500 shares; (3) orders with the order share that is not a multiple of 500 shares. *Oib_SmallRetail*, *Oib_LargeRetail*, and *Oib_INST* in columns (7)-(9) are order imbalances of small retail investors with account value less than 3 million CNY, large retail investors with account value larger than 3 million CNY and institutional investors. The control variables include the previous day return *Ret*(-1), previous week return *Ret*(-6,-2), previous month return *Ret*(-27,-7), the natural logarithm of market capitalization at the previous month end (*Size*), earnings-to-price ratio at the previous month end (*EP*), turnover defined as the previous month average of daily trading volume divided by shares outstanding (*Turnover*), and the natural logarithm of the Amihud illiquidity ratio defined the previous month average of daily absolute return over daily trading volume (*LogAmihud*). The sample includes A-share stocks listed on a major stock exchange in the Chinese stock market for the period January 2017-May 2020. T-statistics based on Newey-West's standard errors with six lags are in brackets.

Dependent Var	Ret								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Independent Var	Oib_ Bench1	Oib_ Bench2	Oib_ Bench3	Oib_ Improve1	Oib_ Improve2	Oib_ Improve3	Oib_ SmallRetail	Oib_ LargeRetail	Oib_ INST
Oib(-1)	-0.0011***	-0.0005***	0.0001	-0.0012***	0.0000	0.0001**	-0.0015***	0.0005***	0.0007***
	[-12.87]	[-8.66]	[1.43]	[-12.13]	[0.94]	[2.13]	[-21.15]	[10.59]	[19.43]
Ret(-1)	0.0130***	0.0323***	0.0380***	0.0075	0.0377***	0.0372***	0.0092**	0.0370***	0.0296***
	[3.07]	[7.21]	[8.22]	[1.61]	[7.98]	[7.85]	[2.00]	[7.88]	[6.13]
Ret(-6,-2)	-0.0095***	-0.0084***	-0.0084***	-0.0090***	-0.0085***	-0.0082***	-0.0097***	-0.0081***	-0.0084***
	[-4.99]	[-4.38]	[-4.36]	[-4.75]	[-4.41]	[-4.26]	[-5.06]	[-4.19]	[-4.43]
Ret(-27,-7)	-0.0025***	-0.0024***	-0.0025***	-0.0026***	-0.0025***	-0.0024***	-0.0028***	-0.0024***	-0.0026***
	[-3.34]	[-3.16]	[-3.35]	[-3.41]	[-3.31]	[-3.21]	[-3.66]	[-3.18]	[-3.47]
Size	0.0002	0.0002*	0.0002*	0.0001	0.0002*	0.0002*	0.0002*	0.0002*	0.0002*

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	[1.58]	[1.77]	[1.77]	[1.37]	[1.69]	[1.72]	[1.92]	[1.79]	[1.65]
Turnover	-0.0090***	-0.0095***	-0.0092***	-0.0090***	-0.0091***	-0.0090***	-0.0093***	-0.0092***	-0.0092***
	[-2.63]	[-2.79]	[-2.70]	[-2.64]	[-2.68]	[-2.64]	[-2.67]	[-2.70]	[-2.70]
LogAmihud	0.0000	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
	[0.42]	[0.79]	[1.13]	[0.59]	[1.12]	[1.12]	[0.66]	[1.06]	[1.07]
EP	0.0026	0.0030	0.0029	0.0034*	0.0029	0.0030	0.0032	0.0029	0.0031
	[1.36]	[1.52]	[1.49]	[1.77]	[1.48]	[1.56]	[1.61]	[1.52]	[1.58]
Adj.Rsq	0.0862	0.0846	0.0833	0.0863	0.0833	0.0834	0.0886	0.0834	0.0847

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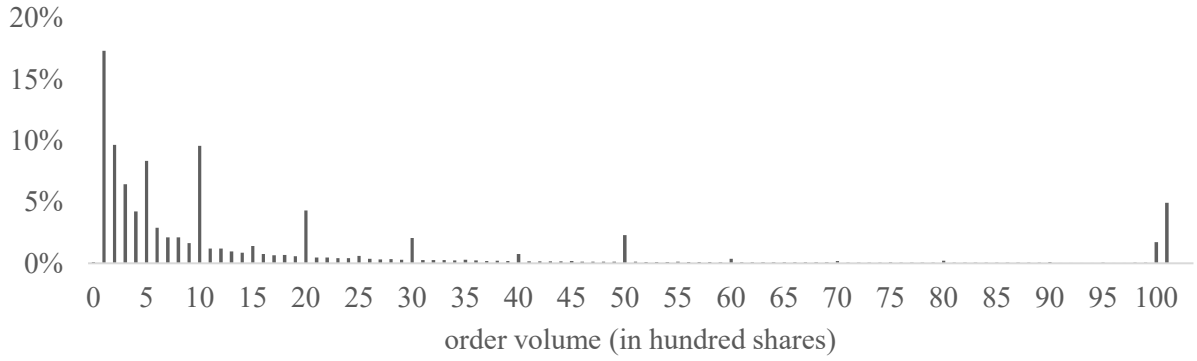
Figure I. Order Share Roundedness

Panel A shows the proportion of order volume from 100 shares to 10,000 shares. Panel B shows the correlation of proportion of order share and proportion of investor order by order volume:

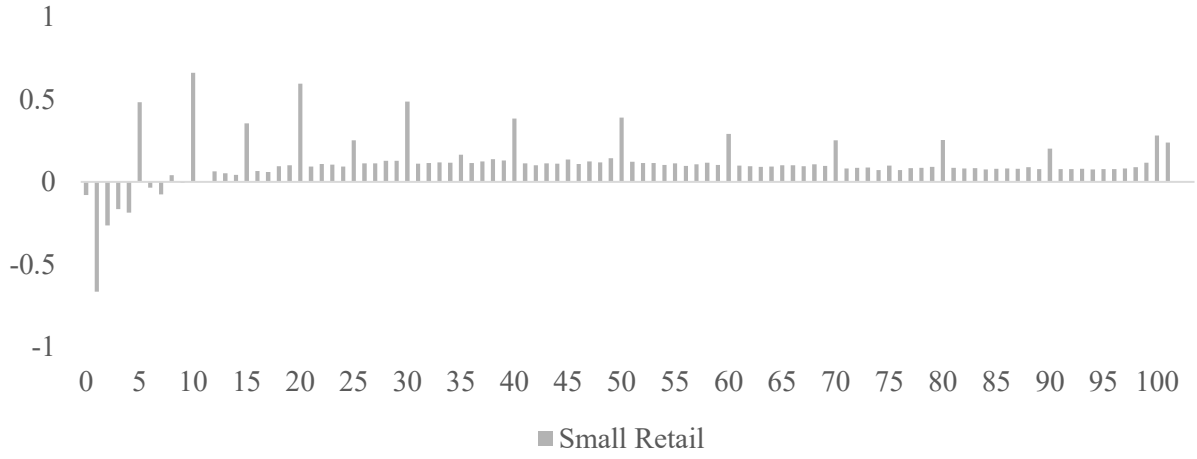
$$Corr(\frac{OrderNum_{i,t,V}}{\sum_V OrderNum_{i,t,V}}, \frac{OrderNum_{i,t,G}}{\sum_G OrderNum_{i,t,G}}),$$

where $OrderNum_{i,t,V}$ is the number of orders of stock i and day t for order share V, $OrderNum_{i,t,G}$ is the number of orders of stock i and day t for investor group G. *SmallRetail* includes small retail investors with account size less than 3 million CNY; *LargeRetail* includes large retail investors with account size larger than 3 million CNY; *INST* includes institutional investors. The sample includes A-share stocks listed on a major stock exchange in the Chinese stock market for the period January 2017-May 2020.

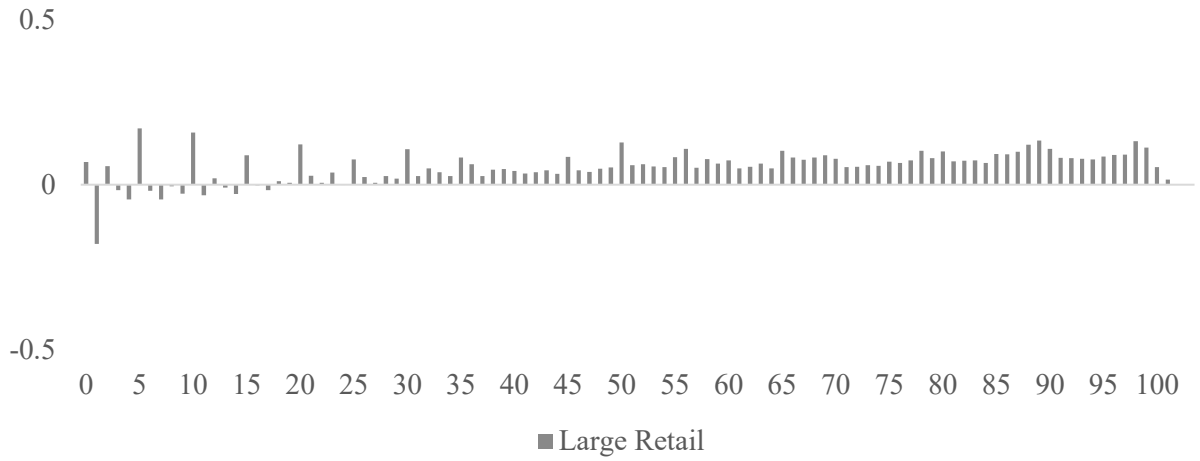
Panel A. Proportion of order volume group

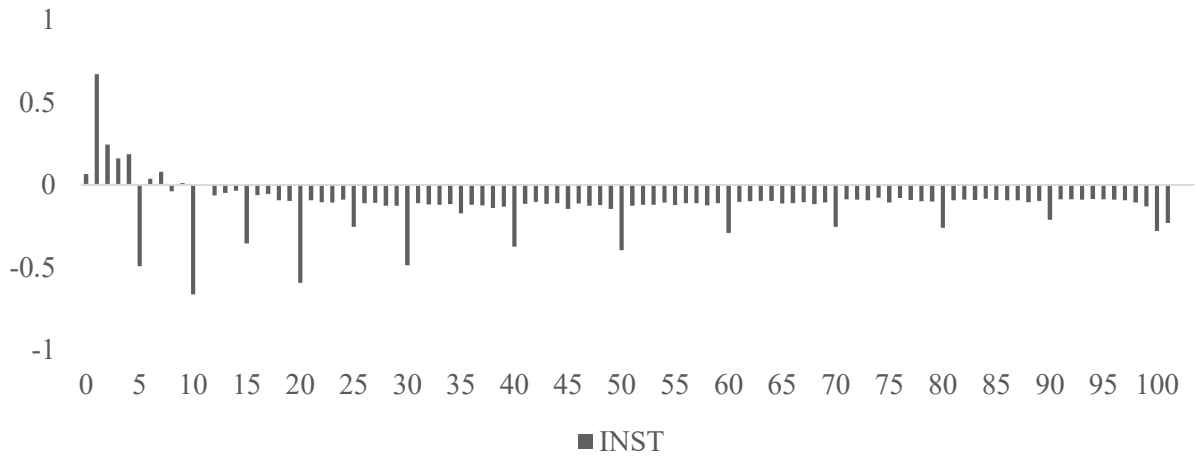


Panel B. Correlation of order share proportion and investor order proportion by order volume (in hundred shares)



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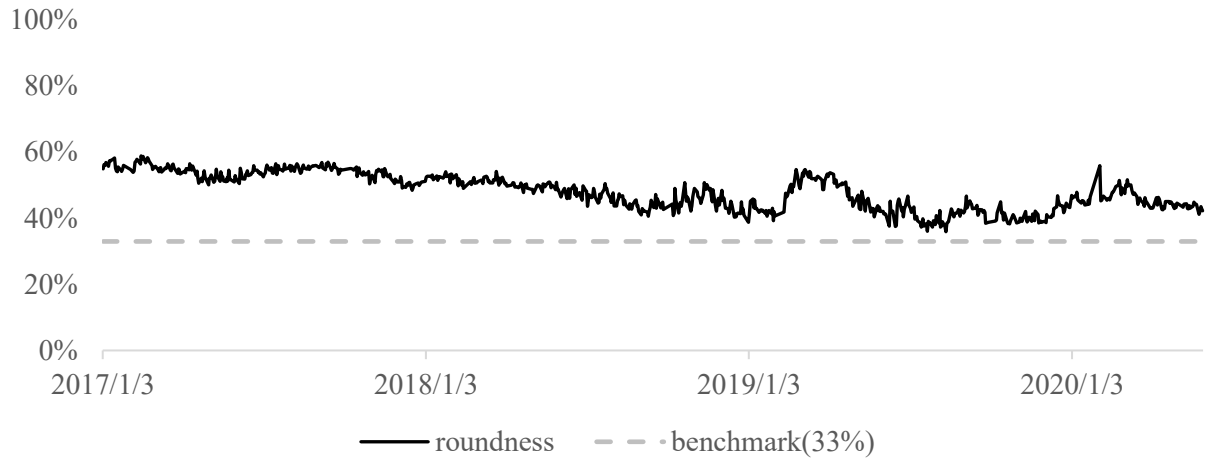
Figure II. Time series of order share roundedness

Panel A presents the time series of cross-sectional average of order share roundedness. Order share roundedness for stock *i* and day *t* is the percentage of orders in multiples of 500 shares out of the sum of multiples of 500 shares and all orders that are either 100 shares more or less than multiples of 500 shares:

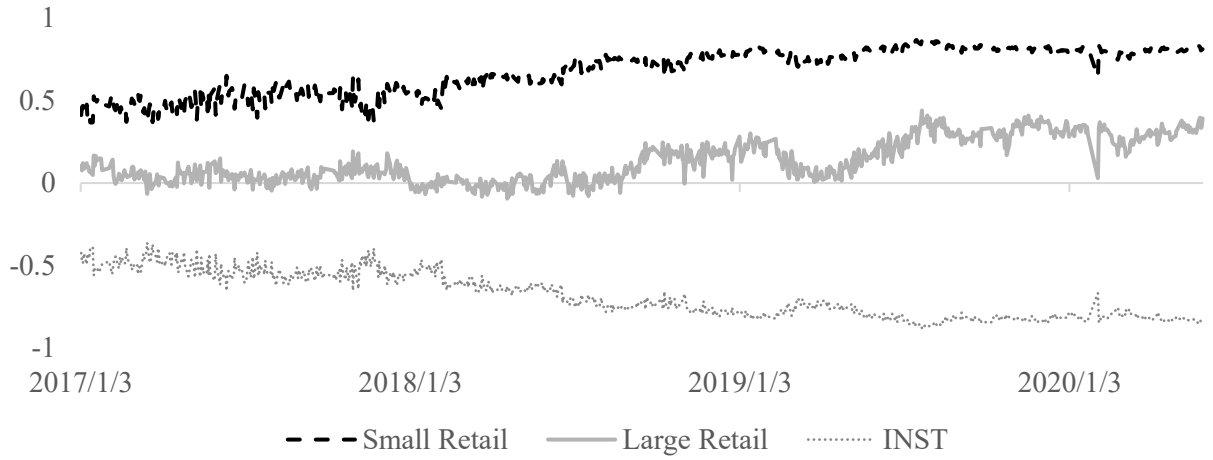
$$Roundedness_{i,t} = \frac{\#\ Orders\ in\ multiples\ of\ 500\ shares_{i,t}}{\#\ Orders\ in\ multiples\ of\ 500\ shares_{i,t} + \#\ Orders\ in\ multiples\ of\ 500\ shares\ \pm\ 100\ shares_{i,t}}$$

Panel B shows the time series of correlation of order share roundedness and investor order proportion, $Corr(Roundedness_{i,t}, \frac{OrdNum_{i,t,G}}{\sum_G OrdNum_{i,t,G}})$, where $OrderNum_{i,t,G}$ is the number of orders of stock *i* and day *t* for investor group *G*. The sample includes A-share stocks listed on a major stock exchange in the Chinese stock market for the period January 2017-May 2020.

Panel A. Time series of order share roundedness



Panel B. Time series of correlation of order share roundedness and investor order proportion

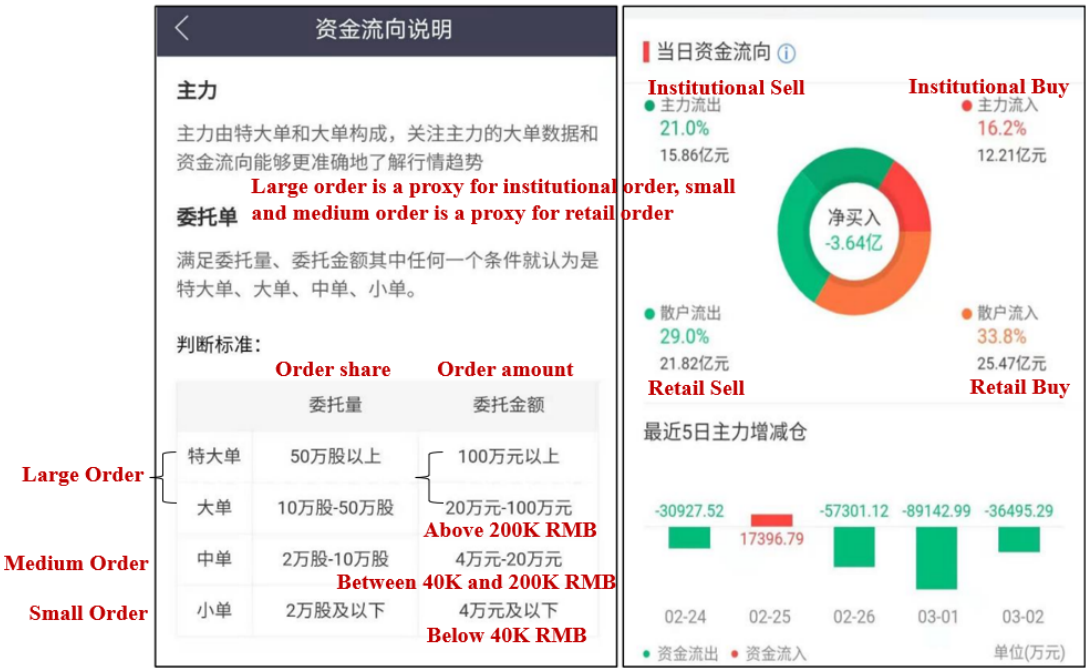


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Internet Appendix

Appendix A. Order Categorization in the Industry

This illustration depicts how orders are categorized as retail and institutional orders based on size in the industry, along with the order flow data for each size group presented on financial information provider applications and platforms.



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Appendix B. Institutional Details of Chinese Stock Market

Stock exchanges in China are order-driven markets without designated market makers, using an electronic automated trading system. There are two types of orders: limit orders and market orders. In a market order, investors specify the quantity only, and the order is immediately executed at the prevailing market price. In a limit order, investors set both the quantity and price, and the order is executed at the limit price or lower (or higher) for a limit buy (sell) order. Limit orders are stored in the limit order book, and the top five bid and ask prices and the corresponding depths are disseminated continuously to the public. Buy and sell orders are matched continuously, following the principle of price and time priority for transactions. Unexecuted limit orders expire at the end of the trading day. The tick size (minimum price variation unit) is 0.01 CNY, and the minimum unit of orders is 100 shares (one lot).

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Appendix Table I. Previous Studies on Tracking Retail and Institutional Trades

Publication Details	Data	Research Questions and Main Findings
Lee and Radhakrishna, Journal of Financial Markets, 2000. "Inferring investor behavior: Evidence from TORQ data."	U.S. public data, the TORQ dataset which contains trades, quotes, order processing, and audit trail data for a sample of 144 NYSE stocks from November 1990 to January 1991.	Research Question: Evaluate the Lee-Ready (1991: Journal of Finance 46, 733-746) algorithm for distinguishing trade direction, and examine the use of trade size as a proxy for trader identity. Findings: (1) For those trades that can be classified, the Lee-Ready algorithm is 93% accurate. (2) Construct a firm-specific trade size proxy that is highly effective in separating the trading activities of individual and institutional investors.
Hvidkjaer, Review of Financial Studies, 2008. "Small Trades and the Cross-Section of Stock Returns."	U.S. public data, transactions data from TAQ and ISSM from January 1983 to December 2005.	Research Question: Use volume arising from small trades to analyze the effect of retail investor trading behavior on the cross-section of stock returns. Findings: (1) Stocks with intense sell-initiated small-trade volume, measured over the past several months, outperform stocks with intense buy-initiated small-trade volume.
Barber, Odean, and Zhu, Review of Financial Studies, 2008. "Do retail trades move markets?	U.S. public data, tick-by-tick trades from TAQ and ISSM over 1983 to 2001.	Research Question: The trading of individual investors and stock returns. Findings: (1) Small trade order imbalance correlates well with order imbalance based on trades from retail brokers. (2) Retail investors herd. (3) Small trade order imbalance negatively forecasts future annual returns. (4) Small trade order imbalance positively predicts future week returns.
Campbell, Ramadorai, and Schwartz, Journal of Financial Economics, 2009. "Caught on tape: Institutional trading, stock	U.S. public data, tick-by-tick trades from TAQ from January 1993 to December 2000.	Research Question: Infer daily institutional trading behavior from TAQ. Findings: (1) Daily institutional trades are highly persistent and respond positively to recent daily returns but negatively to longer-term past daily returns.

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returns, and earnings announcements."		(2) Institutional trades, particularly sells, appear to generate short-term losses but longer-term profits. (3) One source of these profits is that institutions anticipate both earnings surprises and post-earnings announcement drift.
Kaniel, Saar, and Titman, Journal of Finance, 2008. "Individual investor trading and stock returns."	U.S. proprietary data, all retail orders from NYSE CAUD file from year 2000 to 2003.	Research Question: The relation between net individual investor trading and short horizon cross-sectional stock returns. Findings: (1) Individuals buy stocks following declines in the previous month and sell following price increases. (2) Positive excess returns in the month following intense buying by individuals and negative excess returns after individuals sell.
Cready, Kumas, and Subasi, Journal of Accounting Research, 2014. "Are Trade Size-Based Inferences About Traders Reliable? Evidence from Institutional Earnings-Related Trading."	U.S. proprietary data, daily institutional trading data from Ancerno Ltd. from 2003 to 2010.	Research Question: Assess the degree to which the misclassifications give rise to spurious inferences about "small" and "large" investor trading activities. Findings: (1) These institutions are heavily involved in small transaction activity. (2) They increase their order sizes substantially in announcement periods relative to nonannouncement periods.
Barrot, Kaniel, and Sraer, Journal of Financial Economics, 2016. "Are retail traders compensated for providing liquidity?"	French proprietary data, from a leading European online broker between 2002 and 2010.	Research Question: Whether individual investors provide liquidity to the stock market and whether they are compensated for doing so. Findings: (1) The ability of aggregate retail order imbalances to predict short-term future returns is significantly enhanced during times of market stress. (2) Individual investors do not reap the rewards from liquidity provision because they experience a negative return on the trading day and reverse their trades for too long after the liquidity provision dissipated.
Boehmer, Jones, Zhang and Zhang, Journal of Finance, 2021. "Tracking retail investor activity."	U.S. public data, an algorithm to identify marketable retail trades from TAQ over year 2010 to 2015.	Research Question: Provide an algorithm to identify marketable retail purchases and sales using TAQ. Findings: (1) Provide and validate the algorithm.

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		(2) Marketable retail order imbalance positive predict future weekly stock return. (3) Predictive power of marketable retail order imbalance could be attributable to order flow persistence, contrarian trading, public news sentiment and unexplained part.
Barber, Huang, Odean, and Schwarz, Journal of Finance, 2022. "Attention-Induced Trading and Returns: Evidence from Robinhood Users."	U.S. public data, Robinhood investors holding and retail trades identified by Boehmer et al. (2021) from TAQ from May 2018 to August 2020.	Research Question: The influence of financial innovation by fintech brokerages on individual investors' trading and stock prices. Findings: (1) Robinhood investors engage in more attention-induced trading than other retail investors. (2) Intense buying by Robinhood users forecasts negative returns. Average 20-day abnormal returns are −4.7% for the top stocks purchased each day.
Farrell, Green, Jame and Markov, Journal of Financial Economics, 2022. "The democratization of investment research and the informativeness of retail investor trading."	U.S. public data, retail trades identified by Boehmer et al. (2021) from TAQ from 2006 to 2017.	Research Question: The effects of social media on the informativeness of retail trading. Findings: (1) The ability of retail order imbalances to predict the cross-section of stock returns and cash-flow news increases sharply in the intraday post-publication window relative to the pre-publication window.
Jiang, Liu, Peng, and Wang, Review of Finance, 2022. "Investor Attention and Asset Pricing Anomalies."	Chinese public data, transaction-level data for stocks listed on the Shanghai Stock Exchange between April 15 2013 and December 31 2015, from RESSET.	Research Question: The relationship between investor attention and financial market anomalies. Findings: (1) Anomaly returns tend to be higher following high-attention days. (2) Large traders trade on anomaly signals more aggressively upon observing higher attention.
Ye, Wang, Wan and Yang, Journal of Management Sciences in China, 2020. "Intolerable leakage: Informed trading before cuts on	Chinese public data, transaction data from Wind from 2014 to 2015.	Research Question: The trading dynamics in Chinese stock markets before the PBC's interest rate cuts and reserve requirements rate cuts. Findings:

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interest rate and reserve requirement ratio in 2015.”		(1) The return by opening price, volume per trade and the probability of informed trading appear a significant growth, whereas the volatility and the number of trades dramatically drop. (2) The volume and proportion of the big trades rise significantly, and the probability of informed trading both in big and small trades is also higher than in the normal situations.
Xu, Xu, and Zhu, Journal of Management World, 2022. “The "Large-Volume Trading Anomaly" in China's A-Share Market”	Chinese public data, transaction data from a brokerage and from Resset from January 2006 to February 2019.	Research Question: The role of large-volume trades on the pricing of stock returns. Findings: (1) The long-short portfolio that buys the top decile (high large-volume trading index) and sells the bottom (low large-volume trading index) decile achieves an annualized abnormal return of 20%. (2) The large volume trading anomaly is more prevalent in stocks that are smaller, covered by fewer financial analysts, and faced with severer short- selling constraints. (3) Herding and overreaction of retail investors are the important reasons in explaining the large-volume trading anomaly.
An, Lou, and Shi, Journal of Monetary Economics, 2022. “Wealth redistribution in bubbles and crashes.”	Chinese proprietary data, account trading from a major stock exchange in China, 2014-15 bubbles and crashes episode	Research Question: What are the social-economic consequences of financial market bubbles and crashes? Findings: (1) The largest 0.5% households in the equity market gain, while the bottom 85% lose, 250B CNY through active trading in this period.
Jones, Shi, Zhang, and Zhang, Journal of Financial and Quantitative Analysis, 2023. “Retail Trading and Return Predictability in China.”	Chinese proprietary data, account trading and holding from a major stock exchange in China between January 2016 and June 2019	Research Question: Heterogeneity in trading dynamics and performances of Chinese retail investors. Findings: (1) Retail investors with smaller account sizes cannot predict future returns correctly, display daily momentum patterns, fail to process public news, and show overconfidence and gambling preferences.

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		(2) retail investors with larger account balances predict future returns correctly, display contrarian patterns, and incorporate public news in trading. (3) smaller retail investors suffer from poor stock selection abilities and trading costs, while large retail investors' stock selection abilities are offset by trading costs.
Chen, Yuan, and He, Economic Research Journal, 2013. "The Study on Behavior of Heterogeneous Investor and Price Information."	Chinese proprietary data, investor group level trading from Shenzhen Stock Exchange from 2006 to 2010.	Research Question: The dynamic behavior of security prices in the presence of heterogeneous investors. Findings: (1) Heterogeneous investors have different behavior in Chinese Stock Market. (2) When small investors buy from other types of investors, the price will drop. On the contrary, the price will rise. (3) Big individual with information advantage and superior ability buys the undervalue stock from small individual who is liquidity provider with psychological bias, then sells overvalue stock back to them.

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